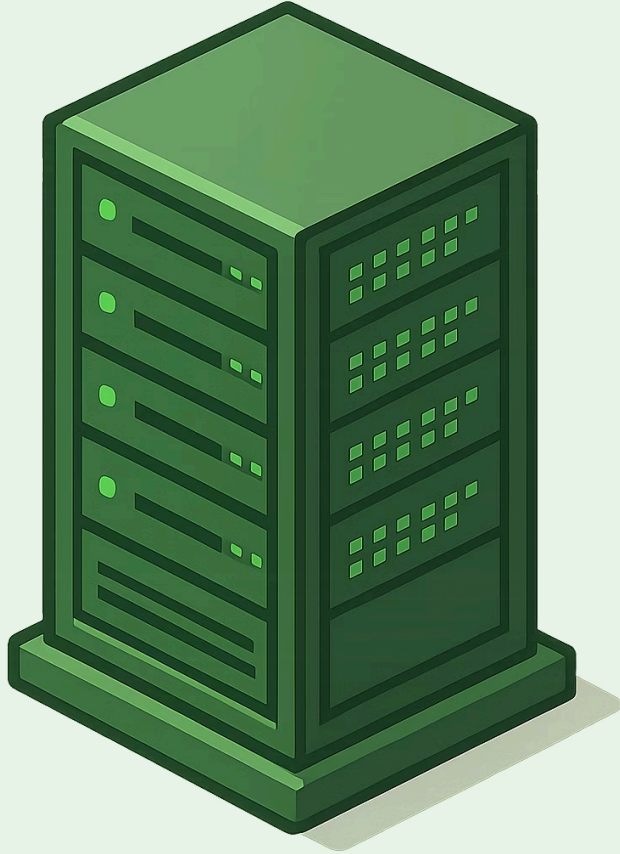




# OpenStack Glance: The Image Service

A practical guide to VM image management, architecture, and operations for cloud engineers and OpenStack operators.

# What is Glance?

## The Role of Glance

Glance is the **Image Service** of OpenStack – the central repository for all Virtual Machine (VM) images in your cloud. It provides a unified REST API for image upload, discovery, retrieval, and lifecycle management.

Every VM boot in OpenStack traces back to Glance. Nova queries Glance to retrieve the appropriate image before handing it off to the hypervisor.

## Core Capabilities

- **Centralized image store**  
Single source of truth for all VM images across the cloud.
- **REST API interface**  
Full CRUD operations via standard HTTP endpoints.
- **Multi-format & multi-backend**  
Supports diverse image formats and pluggable storage backends.
- **Keystone integration**  
All image operations are authenticated and authorized via Keystone tokens.

# Glance Architecture

Glance is composed of three primary layers that work together to handle image management at scale.

## Glance API

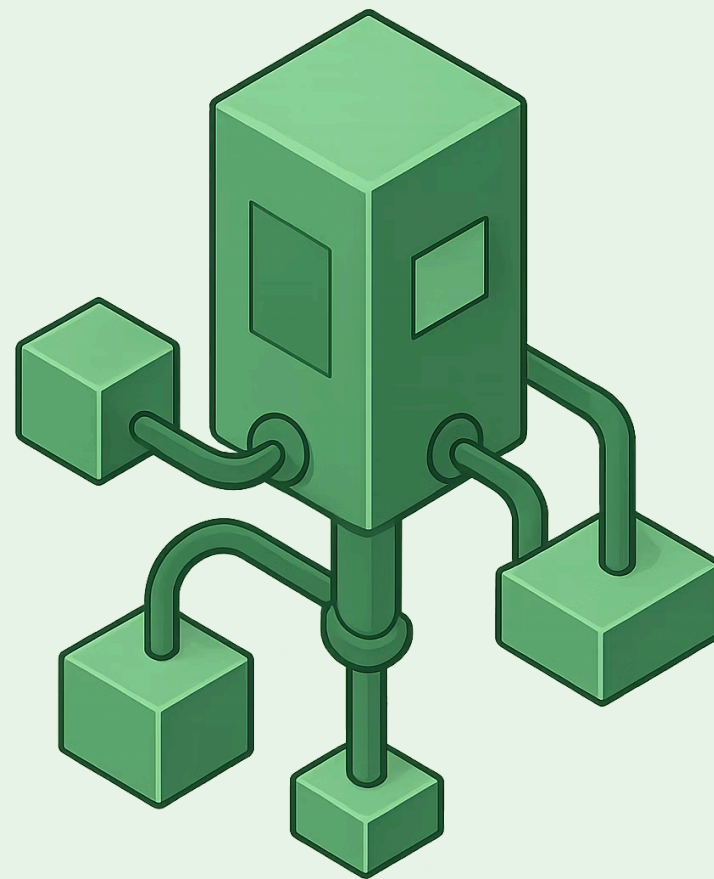
Receives all inbound image management requests from Nova, Horizon, and CLI clients. Validates tokens with Keystone before processing any operation. All communication is REST-based over HTTP/HTTPS.

## Metadata Database

Stores structured image metadata – name, owner, visibility, checksum, format, size, and custom properties. Backed by a relational database (typically MySQL or PostgreSQL) and queryable via API filters.

## Image Backend

Stores the actual binary image files. The backend is configurable – ranging from local filesystem to distributed object stores like Ceph RBD or Swift. Nova accesses images here during VM provisioning.



# Supported Image Formats

Glance supports multiple disk image formats. Choosing the right format depends on your hypervisor, performance requirements, and migration source.



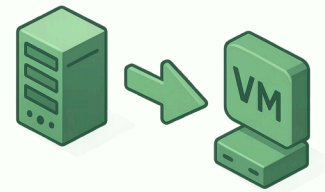
## QCOW2 — KVM/QEMU Default

- Thin provisioning saves disk space
- Native snapshot support built-in
- Smaller footprint vs. RAW
- Recommended for most OpenStack deployments



## RAW — Maximum Performance

- Uncompressed, flat disk image
- Lowest I/O overhead at runtime
- Ideal for Ceph RBD backends
- Higher storage consumption trade-off



## VMDK — VMware Compatible

- VMware virtual disk format
- Common in VMware-to-OpenStack migrations
- Convert using `qemu-img` before upload
- Enables cross-hypervisor portability

 Use `qemu-img convert -f vmdk -O qcow2 source.vmdk dest.qcow2` to convert images before uploading to Glance.

# Image Metadata

Every image in Glance carries a rich set of metadata that drives scheduling, security, and deployment decisions across Nova and other OpenStack services.

## Standard Metadata Fields

### Identity

Image Name, Image ID,  
Owner Project

### Visibility

Public / Private / Shared /  
Community – controls  
cross-tenant access

### Format

Disk Format (qcow2, raw,  
vmdk) and Container  
Format (bare, ovf, ova)

### Requirements

Minimum RAM and Disk  
size – enforced by Nova  
during flavor matching

## Extended & Custom Fields

### Architecture

Target CPU architecture:  
x86\_64, aarch64, ppc64le

### Tags

Searchable labels for  
filtering images by  
environment or application  
type

### Custom Properties

Arbitrary key-value pairs  
for cloud-specific  
metadata and automation  
hooks

### Protected Flag

Prevents accidental  
deletion of critical base  
images

# Image Storage Backends

Glance is backend-agnostic – the image binary can be stored in several systems depending on your infrastructure and scale requirements.



## Local Filesystem

Simplest option. Stores images directly on the Glance host disk. Suitable for development and single-node labs only.



## Swift Object Storage

OpenStack-native object store. Scalable and redundant – good fit when Swift is already deployed in the environment.



## Ceph RBD

**Production standard.** Distributed, highly available, and efficient – especially with Copy-on-Write (CoW) for fast VM clones.



## Cinder Block Storage

Stores images as Cinder volumes. Enables bootable volume workflows and persistent root disk scenarios.



## NFS / S3-Compatible

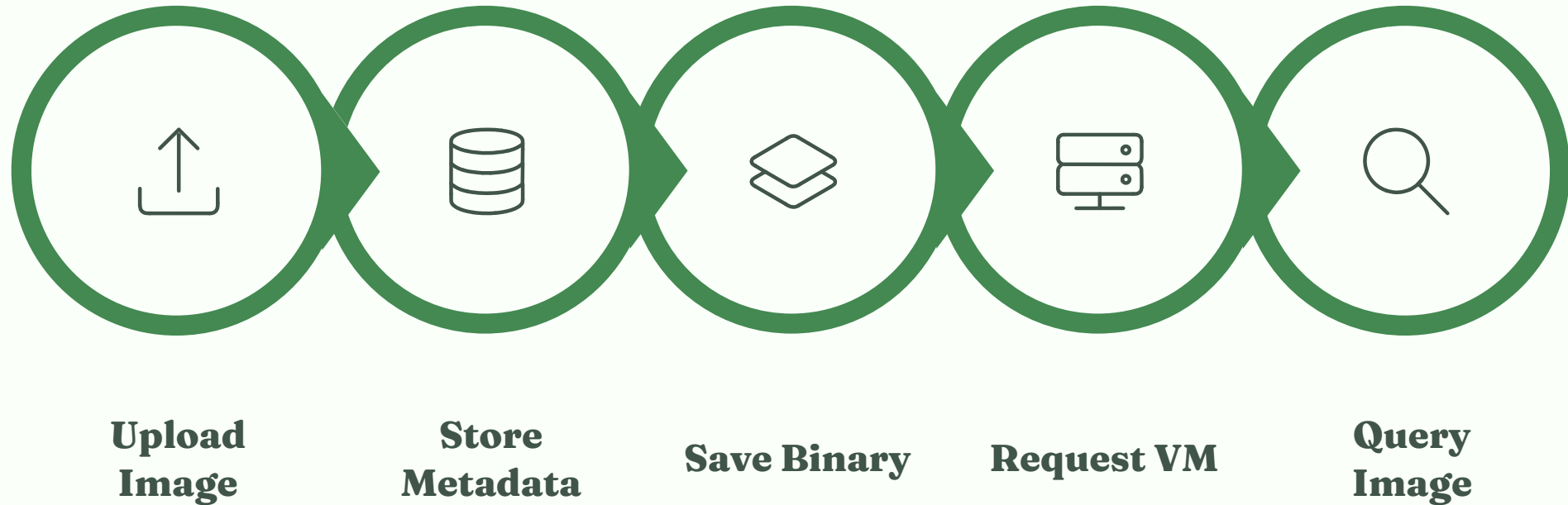
NFS mounts and S3-compatible object stores (via community drivers) extend Glance to hybrid or existing enterprise storage.



**Production Recommendation:** Ceph RBD is the de facto standard for production OpenStack deployments – it delivers scalability, HA, and native CoW cloning that dramatically accelerates VM boot times.

# Glance Workflow: Image Upload to VM Boot

From image upload to a running virtual machine, Glance orchestrates each step of the provisioning pipeline in collaboration with Nova and the hypervisor.



This pipeline ensures that image management is decoupled from compute – Nova never manages image files directly, keeping responsibilities cleanly separated across services.

# Key Features & Enterprise Benefits

Glance is more than a file store – it is a foundational service that enables consistent, scalable, and secure VM provisioning across the entire cloud.



## Centralized & Secure

Single authoritative image source. All access controlled via Keystone RBAC and image visibility policies.



## Faster Provisioning

Reusable golden images eliminate repetitive OS installs. CoW cloning from Ceph brings boot times to seconds.



## Multi-Tenant Sharing

Images can be scoped as public, private, shared with specific projects, or community-visible across the deployment.



## Scalable & HA

Stateless API tier scales horizontally. Combined with Ceph or Swift, the image store is distributed and fault-tolerant.

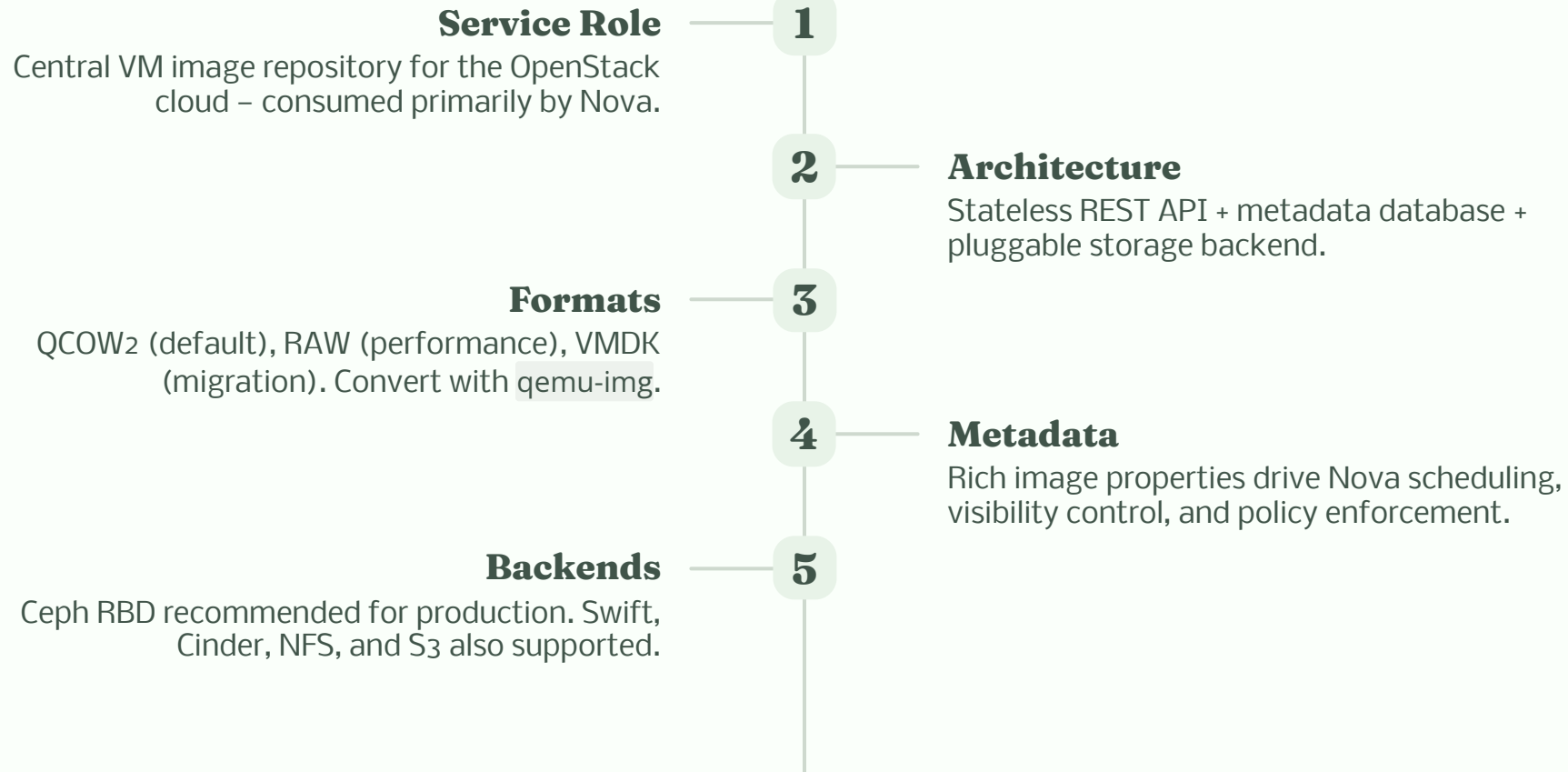


## Deep Integration

Native integration with Nova, Cinder, Horizon, Heat, and Ironic. Metadata drives scheduling, boot, and automation decisions.

# Glance at a Glance

A quick-reference summary of the key concepts covered in this session.



Glance's clean separation of metadata and binary storage makes it resilient, scalable, and straightforward to operate at any cloud scale.

# Further Reading & Next Steps

Ready to go deeper? Explore these practical next steps to strengthen your Glance expertise.

## 1 Explore the Glance API

Practice image CRUD operations using `openstack image CLI` commands and the v2 REST API directly via `curl` or Postman.

## 2 Configure a Ceph RBD Backend

Set up `glance-store` with the RBD driver in a lab environment. Observe how CoW cloning accelerates Nova VM boot sequences.

## 3 Test Image Visibility Policies

Create public, private, and shared images. Verify cross-project access behavior with different Keystone roles and scoped tokens.

## 4 Review Official Documentation

The OpenStack Glance project docs at [docs.openstack.org/glance](https://docs.openstack.org/glance) provide operator guides, configuration references, and upgrade notes.

